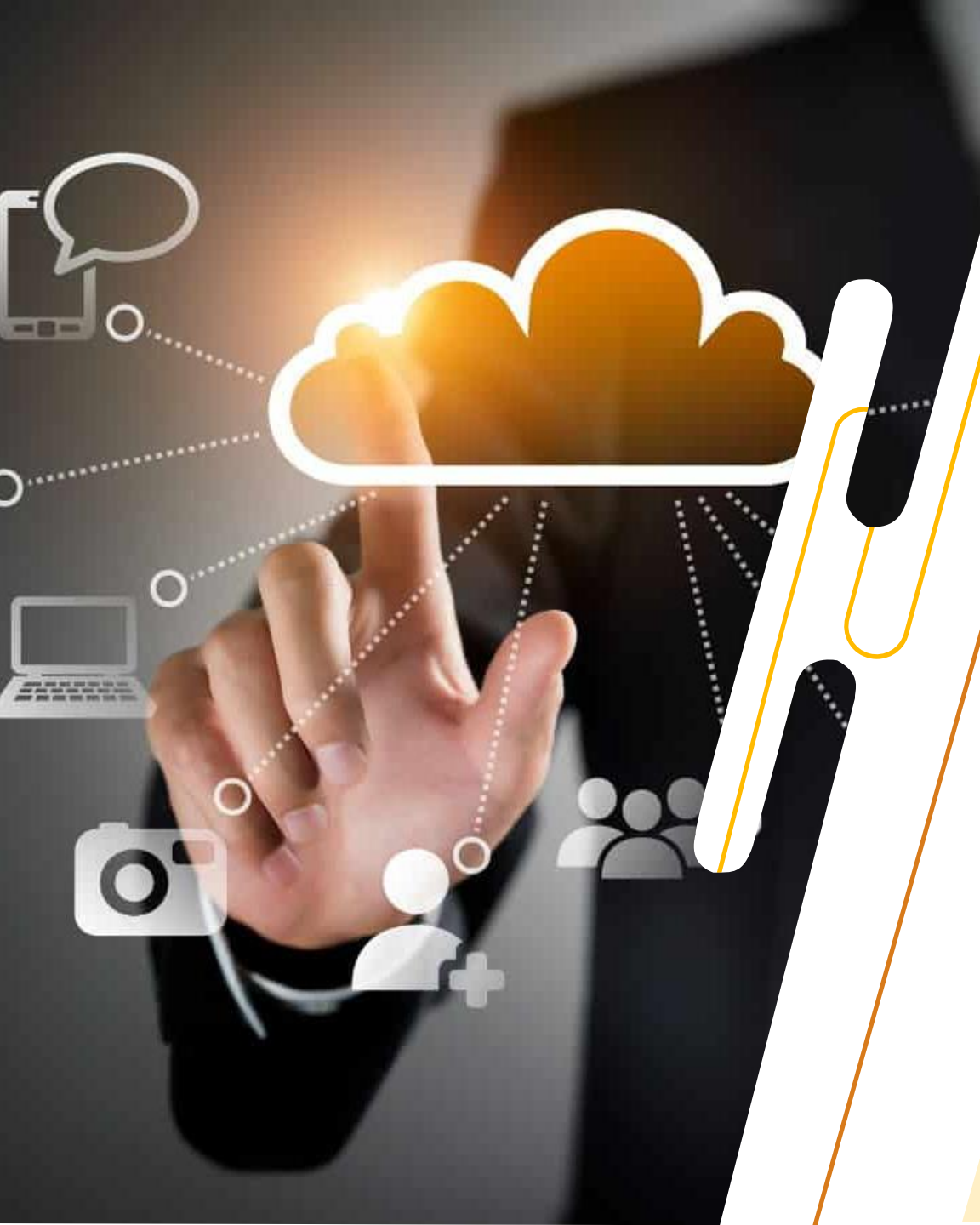




IT3061

Massive Data Processing and Cloud Computing

YEAR 3 SEMESTER 2
DATA SCIENCE SPECIALIZATION

Lecture Outline

- 1. Elastic Load Balancing
- 2. How Elastic Load Balancing works ?
- 3. Types of Load Balancers
- 4. Load Balancer Monitoring
- 5. AutoScaling
- 6. Types of AutoScaling
- 7. How AutoScaling works ?



Elastic Load Balancing

- Distributes incoming application or network traffic across multiple targets in a single Availability Zone or across multiple Availability Zones.
- Scales the load balancer as traffic to the application changes over time.

Application Clients (End Users)



Elastic Load Balancer



Scales & Distributes
Application
Traffic

Application Servers



Load Balancing Algorithms

Static load balancing

Static load balancing algorithms follow fixed rules and are independent of the current server state.

The following are examples of static load balancing.

- Round-robin method
- Weighted round-robin method
- IP hash method

Load Balancing Algorithms

Dynamic load balancing

Dynamic load balancing algorithms examine the current state of the servers before distributing traffic.

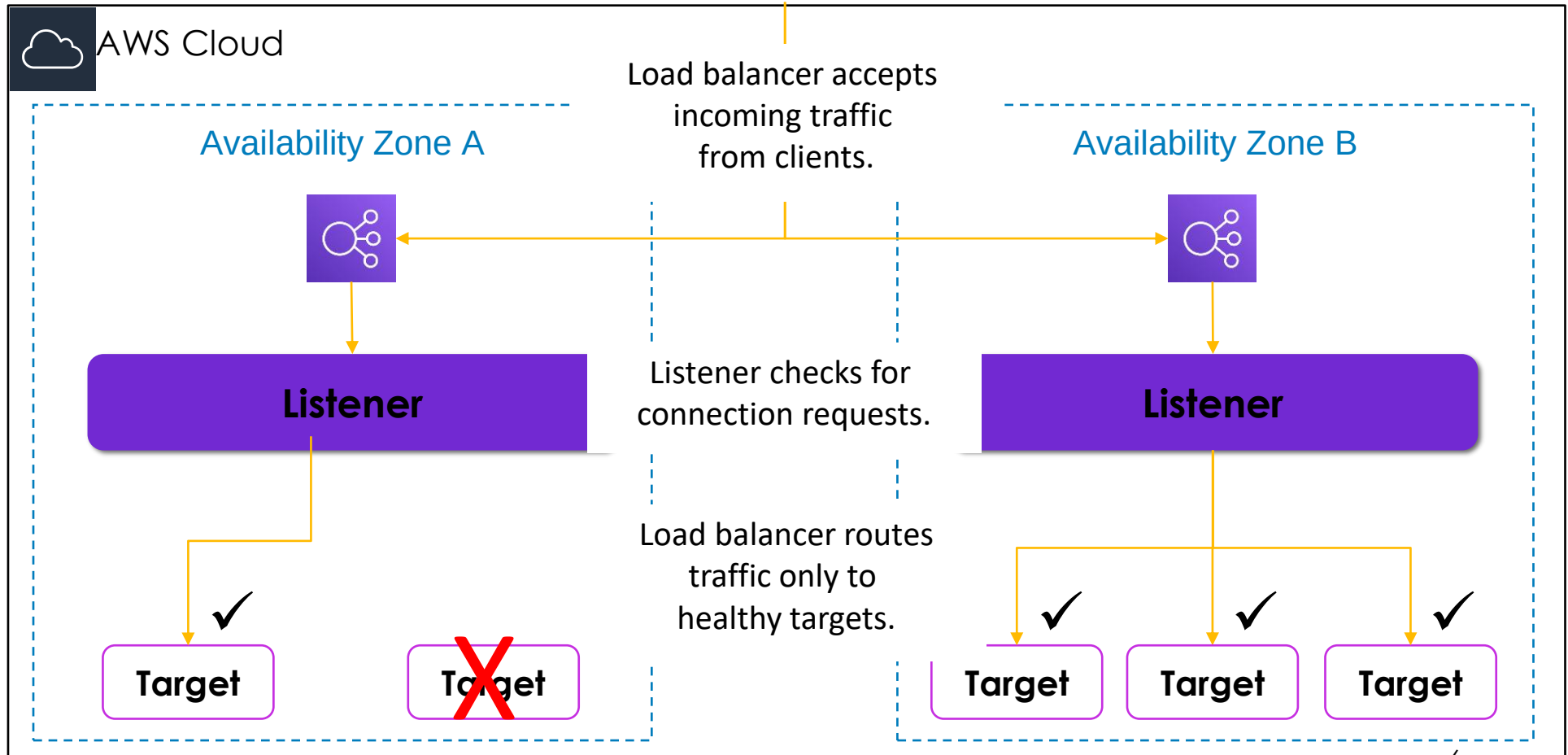
The following are some examples of dynamic load balancing algorithms.

- Least connection method
- Weighted least connection method
- Least response time method
- Resource-based method

How Elastic Load Balancing works ?



Load balancer performs health checks to monitor health of registered targets.



How Elastic Load Balancing works ?

- With Application Load Balancers and Network Load Balancers, you register targets in target groups, and route traffic to the target groups.
- With Classic Load Balancers, you register instances with the load balancer.

Types of Load Balancers - AWS

Application Load Balancer	Network Load Balancer	Classic Load Balancer (Previous Generation)
• Load balancing of HTTP and HTTPS traffic	• Load balancing of TCP, UDP, and TLS traffic where extreme performance is required	• Load balancing of HTTP, HTTPS, TCP, and SSL traffic
• Routes traffic to targets based on content of request • Provides advanced request routing targeted at the delivery of modern application architectures, including microservices and containers	• Routes traffic to targets based on IP protocol data • Can handle millions of requests per second while maintaining ultra-low latencies • Is optimized to handle sudden and volatile traffic patterns	• Load balancing across multiple EC2 instances
• Operates at the application layer (OSI model layer 7)	• Operates at the transport layer (OSI model layer 4)	• Operates at both the application and transport layers.

Types of Load Balancers - Azure

Public Load Balancer	Internal (or Private) Load Balancer
Provide outbound connections for virtual machines (VMs) inside your virtual network	Used in scenarios where private IPs are needed at the frontend only
These connections are accomplished by translating their private IP addresses to public IP addresses	Used to load balance traffic inside a virtual network
Used to load balance internet traffic to your VMs.	Can be accessed from an on-premises network in a hybrid scenario.

Activity : Name the load balancer you would use for the given scenario

You must support traffic to a containerized application.

You have extremely spiky and unpredictable TCP traffic.

You need simple load balancing with multiple protocols.

You need to support a static or Elastic IP address, or an IP target outside a VPC.

You need a load balancer that can handle millions of requests per second while maintaining low latencies.

You must support HTTPS requests.

Activity : Elastic Load Balancing

You must support traffic to a containerized application.	Application Load Balancer
You have extremely spiky and unpredictable TCP traffic.	Network Load Balancer
You need simple load balancing with multiple protocols.	Classic Load Balancer
You need to support a static or Elastic IP address, or an IP target outside a VPC.	Network Load Balancer
You need a load balancer that can handle millions of requests per second while maintaining low latencies.	Network Load Balancer
You must support HTTPS requests.	Application Load Balancer

Load Balancer Monitoring

To use Cloud Services efficiently, you need insight into your cloud resources:

- How do you know when you should **launch more instances / VMs** ?
- Is your **application's performance or availability** being affected by a lack of sufficient capacity ?
- How much of your infrastructure is actually **being used** ?

Load Balancer Monitoring

Following features can be used to monitor load balancers, analyze traffic patterns, and troubleshoot issues with load balancers and targets.

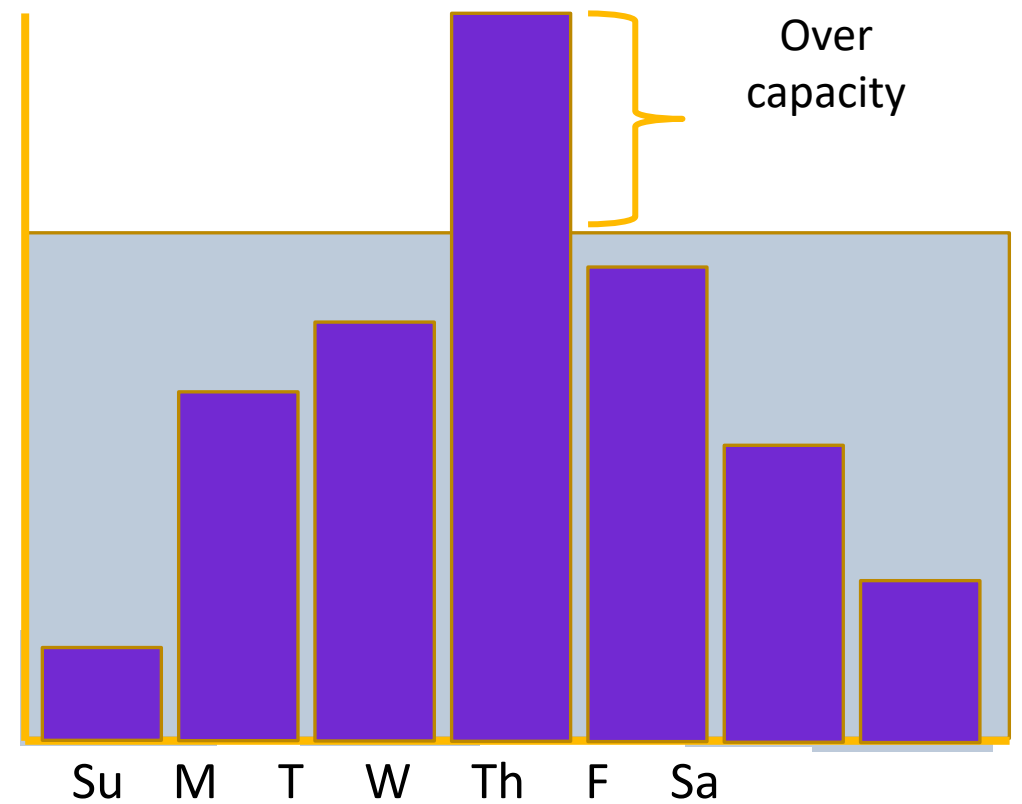
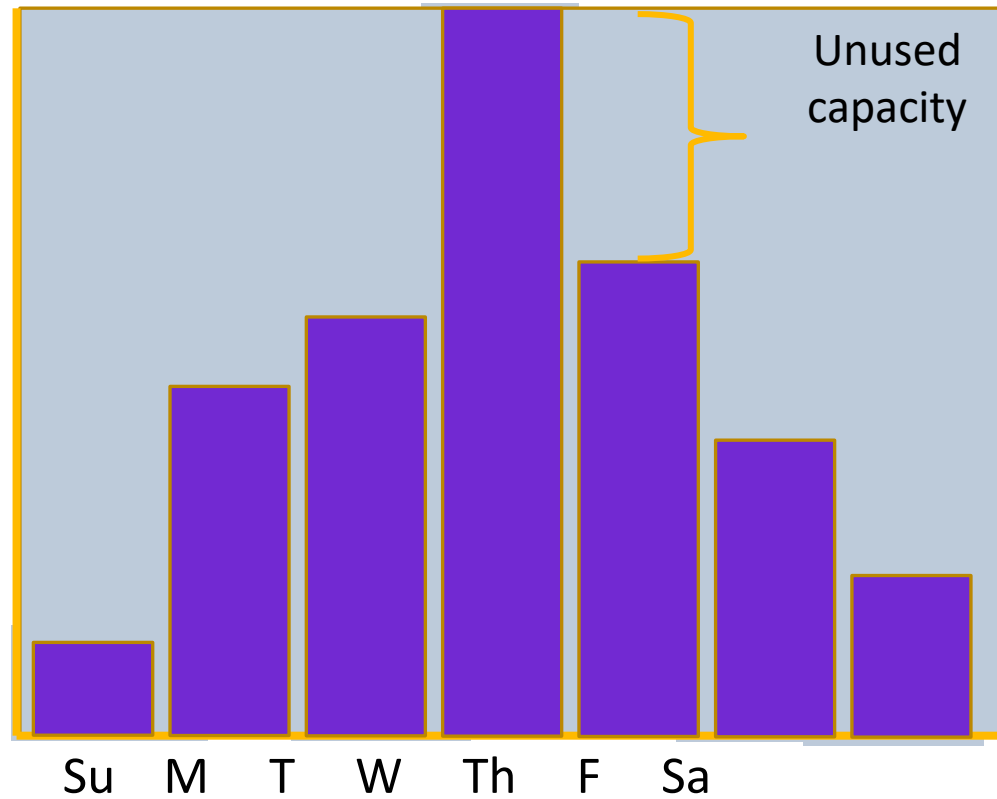
AWS

- CloudWatch metrics
- Access logs
- Connection logs
- Request tracing
- CloudTrail logs

Azure

- Data path availability
- Health probe status
- Utilized SNAT ports
- Provisioning state

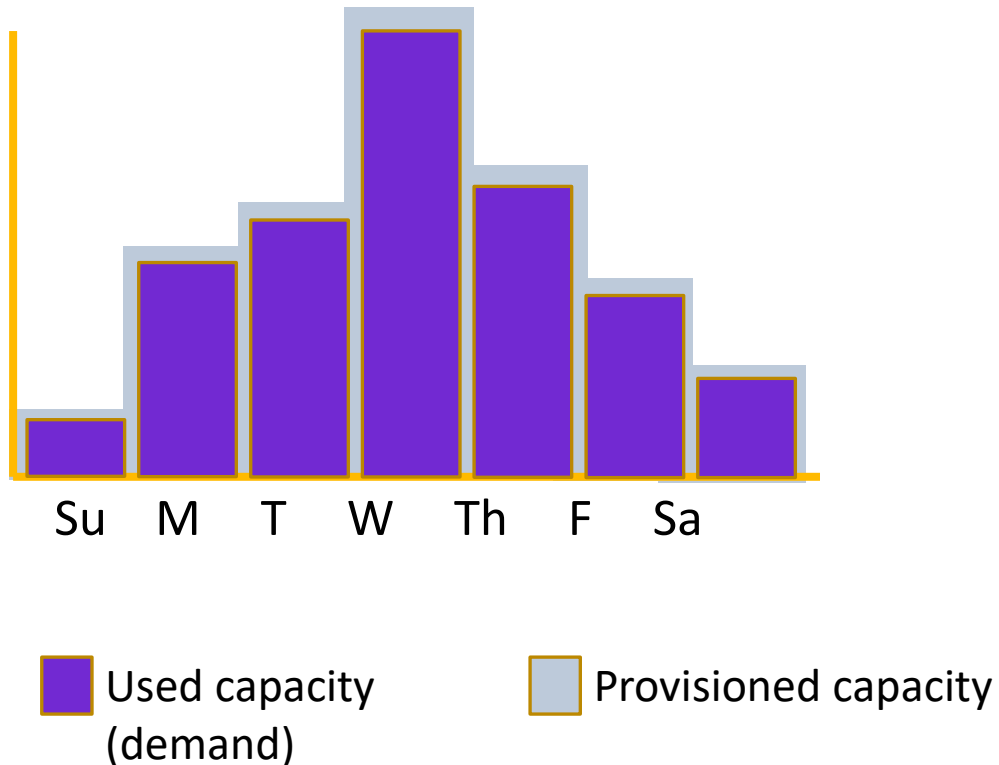
Why Scaling is important ?



Used capacity
(demand)

Provisioned capacity

AutoScaling



- Helps to maintain application availability
- Enables to automatically add or remove resources according to conditions that are define
- Detects impaired instances and unhealthy applications, and replaces the instances without your intervention
- Provides several scaling options – Manual, scheduled, dynamic or on-demand, and predictive

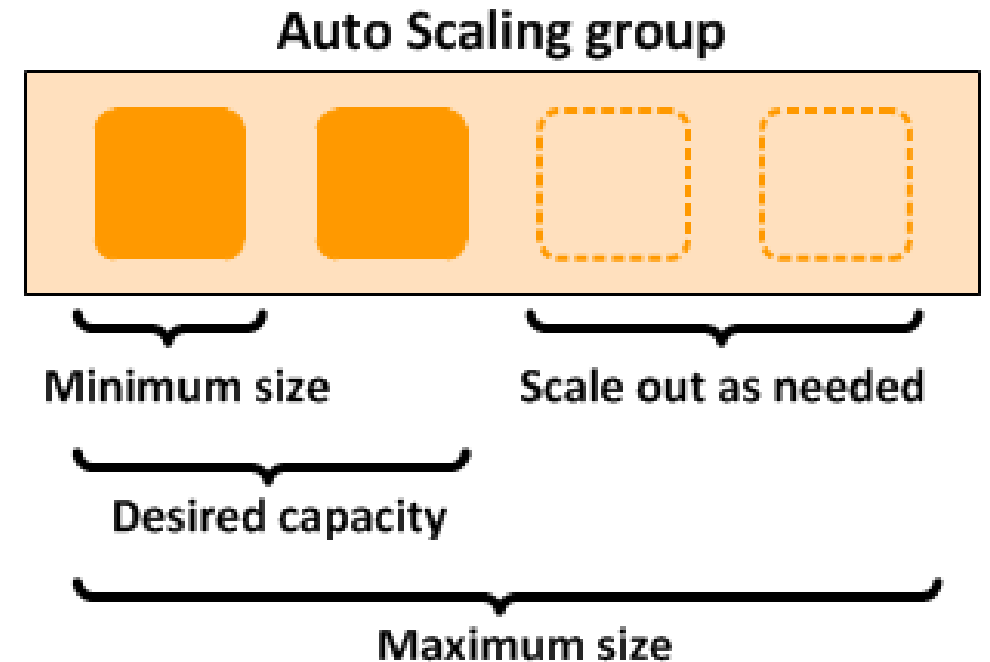
Types of AutoScaling

- **Reactive**

With a reactive autoscaling approach, resources scale up and down as traffic spikes occur.

This approach is closely tied with real-time monitoring of resources.

There is often also a cooldown period involved, which is a set time period where resources stay elevated -- even as traffic drops -- to deal with any additional incremental traffic spikes.



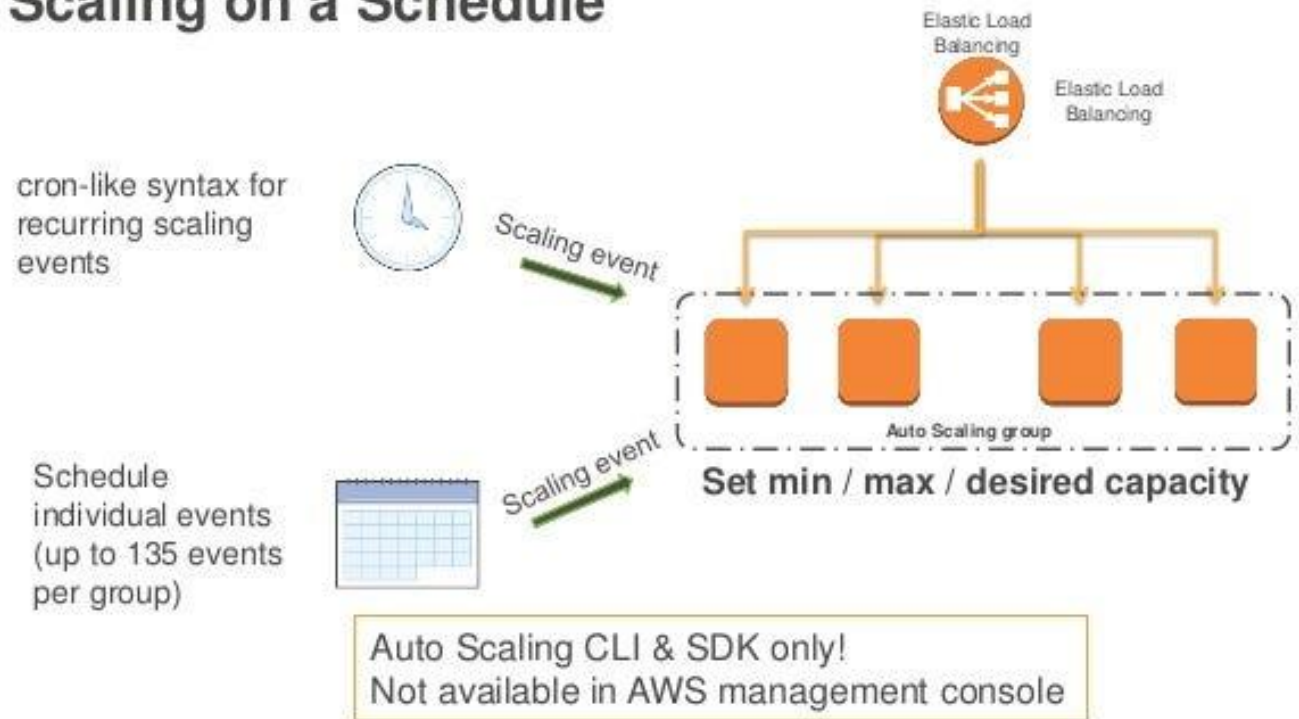
Types of AutoScaling

- **Scheduled (Proactive)**

With a scheduled approach, a user can define a time horizon for when more resources will be added.

For example, ahead of a major event or for a peak time period during the day, rather than waiting for the resources to scale up as demand ramps up, they can be pre-provisioned in anticipation.

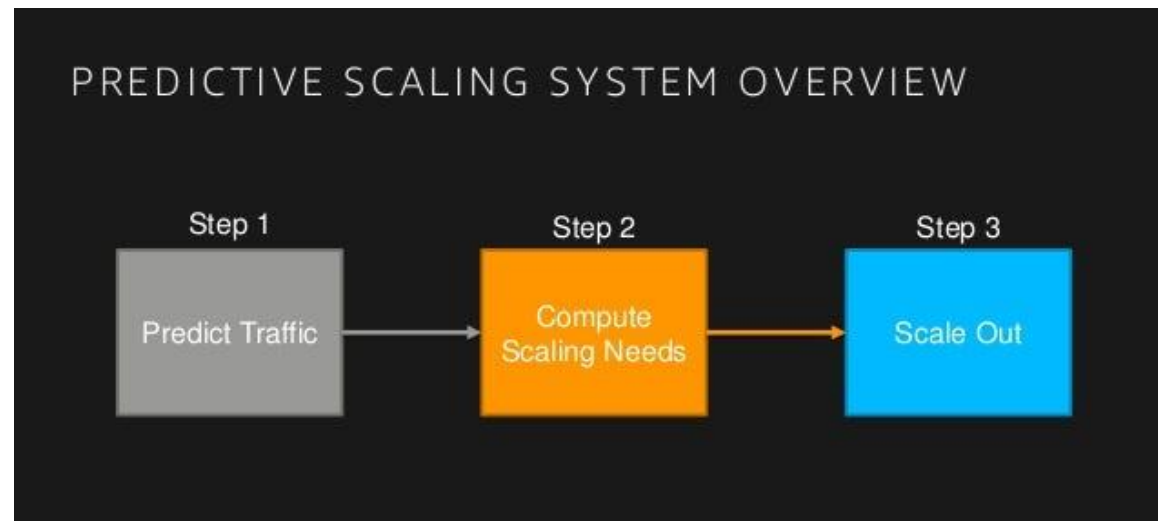
Scaling on a Schedule



Types of AutoScaling

- **Predictive**

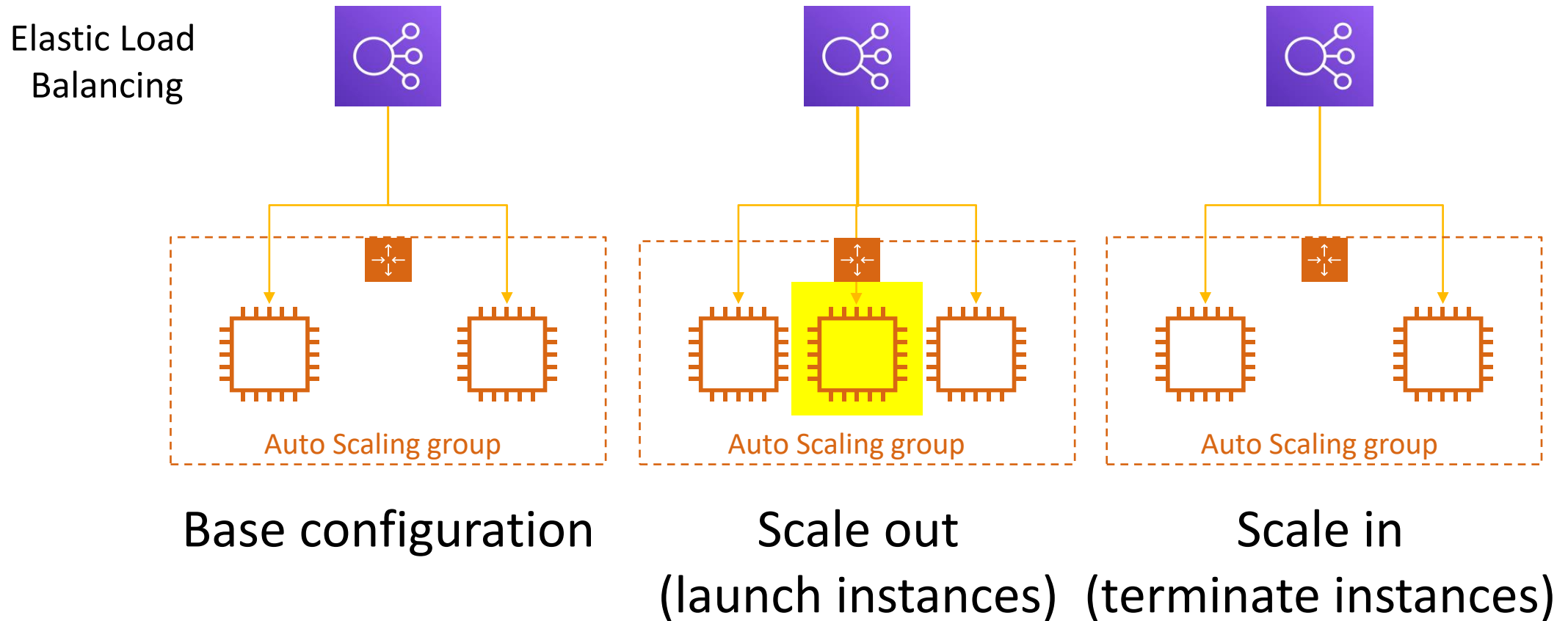
In a predictive autoscaling approach, machine learning and artificial intelligence techniques are used to analyze traffic loads and predict when more or fewer resources will be needed.



Types of Scaling in Cloud

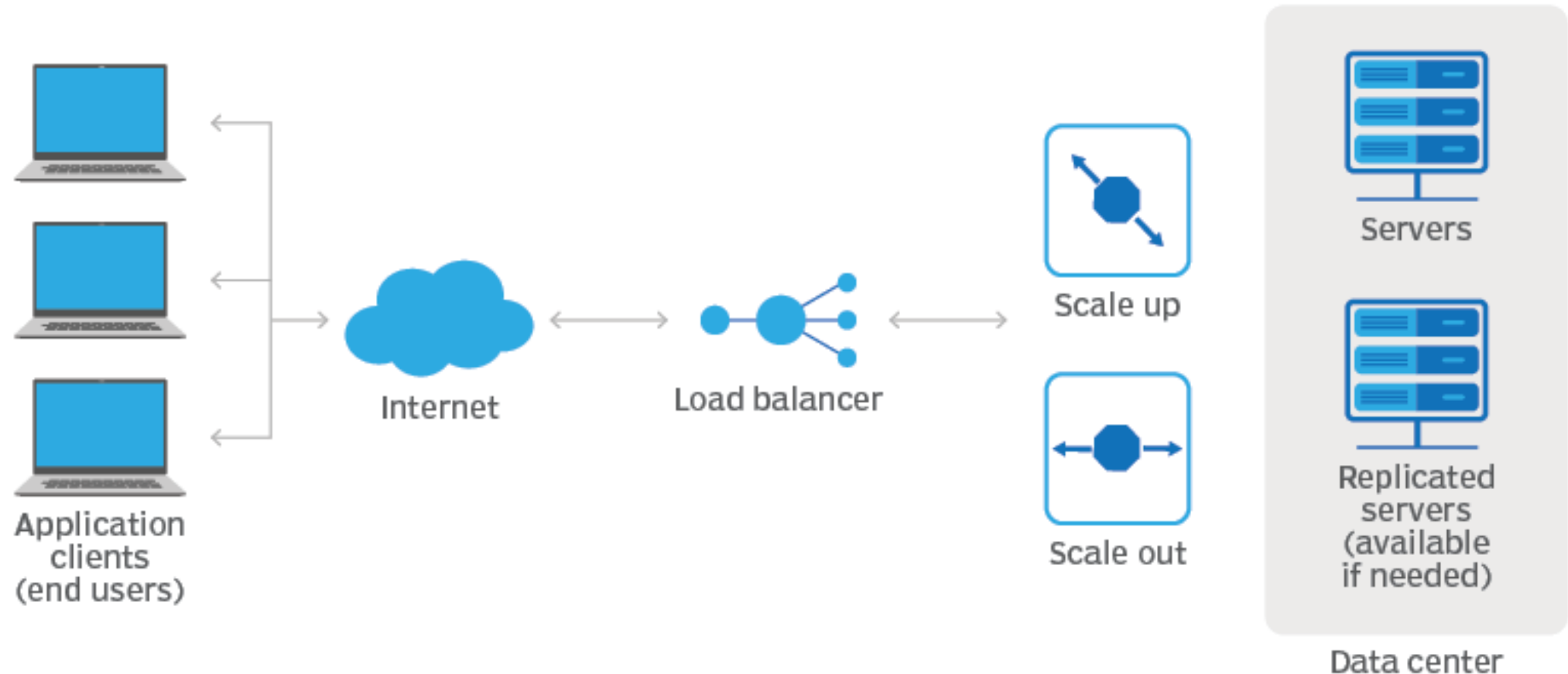
- **Scale up/down (Vertical Scaling)** – adjusting the capacity of an existing resource.
 - scaling up => means enhancing the capabilities of a single machine
 - scaling down => means reducing the capabilities when the load decreases
- **Scale out/In (Horizontal Scaling)** – involves increasing capacity by adding more identical resources to a system
 - scaling in => means reducing capacity by removing nodes or resources
 - scaling out => means increasing capacity by adding nodes or resources

Scaling Out vs Scaling In



An **Auto Scaling group** is a collection of EC2 instances that are treated as a logical grouping for the purposes of automatic scaling and management.

How AutoScaling works ?



Source -

<https://www.techtarget.com/searchcloudcomputing/definition/autoscaling#:~:text=Autoscaling%20provides%20users%20with%20an,spikes%20and%20use%20patterns%20demand.>

AutoScaling Services

- AWS

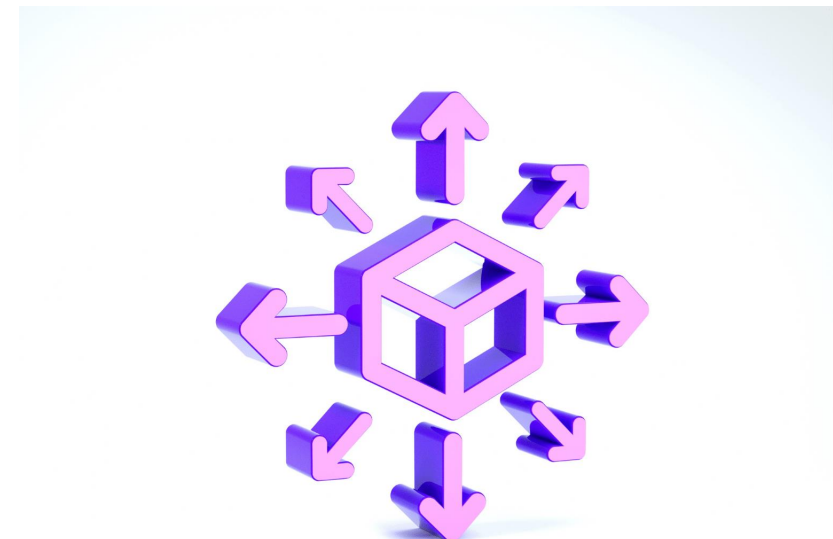
AWS AutoScaling

Amazon EC2 AutoScaling



- Azure

Azure AutoScale

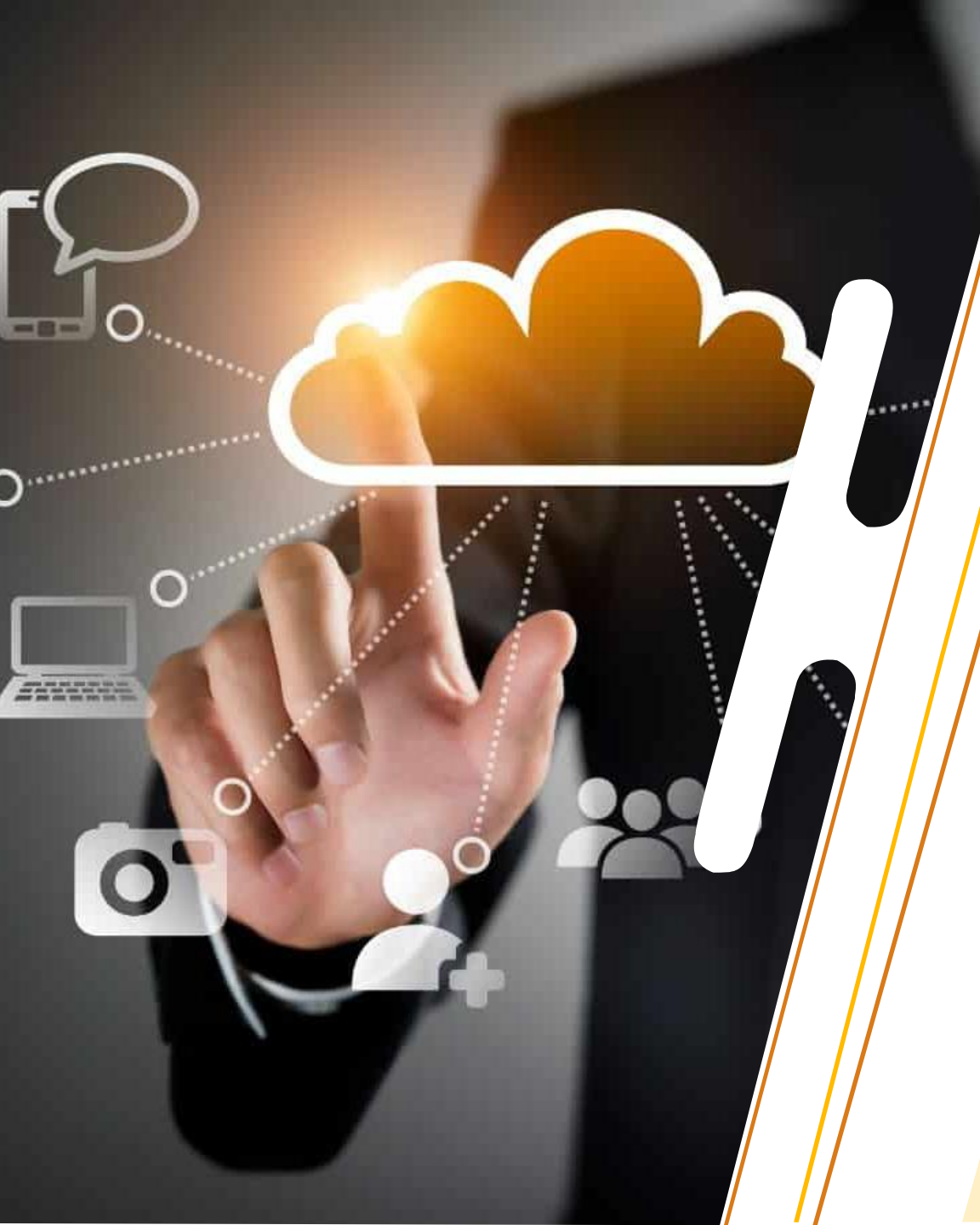




Demo

References

- AWS Academy Cloud Foundations
- AWS Documentation
- Azure Documentation
- <https://medium.com/@damadhav/aws-scaling-reactive-vs-proactive-vs-predictive-2701ad6d48c9>
- <https://www.techtarget.com/searchcloudcomputing/definition/autoscaling#:~:text=Autoscaling%20provides%20users%20with%20an,spikes%20and%20use%20patterns%20demand.>



Next Session :
Load Balancing
and Auto Scaling